Round to ten, hundred and thousand

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Round 468,753 to the nearest 10, 100 and 1,000. Copy or complete the first step.

2. Round 905,449 to the nearest 100.

3. Round 279,650 to the nearest 1,000.

4. Miro says: Miro always rounds up when the number contains a 5 anywhere. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Round 468,753 to the nearest 10, 100 and 1,000. Copy or complete the first step.
   Answer 468,750; 468,800; 469,000.

2. Round 905,449 to the nearest 100.
   Answer 905,400.

3. Round 279,650 to the nearest 1,000.
   Answer 280,000.

4. Miro says: Miro always rounds up when the number contains a 5 anywhere. Is this correct? Explain.
   Answer Only the digit immediately to the right of the rounding place decides the direction.

Round to ten, hundred and thousand

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ______________

1. Round 468,753 to the nearest 10, 100 and 1,000.

2. Round 905,449 to the nearest 100.

3. Round 279,650 to the nearest 1,000.

4. Solve this independently and show every stage: Round 468,753 to the nearest 10, 100 and 1,000.

5. Use a second method or representation for: Round 905,449 to the nearest 100.

6. Check this result using an inverse, estimate or known fact: Round 279,650 to the nearest 1,000.

7. Give three numbers that round to 52,000 to the nearest 1,000.

8. Identify and correct this misconception: Miro always rounds up when the number contains a 5 anywhere.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Round 279,650 to the nearest 1,000.

Teacher answers and checking prompts

1. Round 468,753 to the nearest 10, 100 and 1,000.
   Answer 468,750; 468,800; 469,000.

2. Round 905,449 to the nearest 100.
   Answer 905,400.

3. Round 279,650 to the nearest 1,000.
   Answer 280,000.

4. Solve this independently and show every stage: Round 468,753 to the nearest 10, 100 and 1,000.
   Answer 468,750; 468,800; 469,000.

5. Use a second method or representation for: Round 905,449 to the nearest 100.
   Answer Expected result: 905,400. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Round 279,650 to the nearest 1,000.
   Answer Expected result: 280,000. A valid check must be shown.

7. Give three numbers that round to 52,000 to the nearest 1,000.
   Answer Any three from 51,500 to 52,499.

8. Identify and correct this misconception: Miro always rounds up when the number contains a 5 anywhere.
   Answer Only the digit immediately to the right of the rounding place decides the direction.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Round 279,650 to the nearest 1,000.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Round to ten, hundred and thousand

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ______________

1. Give three numbers that round to 52,000 to the nearest 1,000.

2. Prove the answer to this question, rather than only calculating it: Round 468,753 to the nearest 10, 100 and 1,000.

3. Solve this in two different ways and compare them: Round 905,449 to the nearest 100.

4. Work backwards from the answer to reconstruct the question: 280,000.

5. Explain why this reasoning fails: Miro always rounds up when the number contains a 5 anywhere.

6. Create a new example that tests the same idea as: Round 279,650 to the nearest 1,000.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Give three numbers that round to 52,000 to the nearest 1,000.
   Answer Any three from 51,500 to 52,499.

2. Prove the answer to this question, rather than only calculating it: Round 468,753 to the nearest 10, 100 and 1,000.
   Answer Expected result: 468,750; 468,800; 469,000. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Round 905,449 to the nearest 100.
   Answer Expected result: 905,400. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 280,000.
   Answer One valid reconstruction is: Round 279,650 to the nearest 1,000. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro always rounds up when the number contains a 5 anywhere.
   Answer Only the digit immediately to the right of the rounding place decides the direction.

6. Create a new example that tests the same idea as: Round 279,650 to the nearest 1,000.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.